Challeges:
1. Want sponsorship to occur at same time at each chapter

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Say time committment is up to the person

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If someone is not showing up to required meetings we will remove them

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We 100% encourage peers to not team up with people who aren't as interested as them

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Project 1: Interesting project

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Project 2: Display of competence to the world

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- **First Half:** Build foundation and ask a worthwhile question that you can probably answer
- **Second Half:** Try answering your question

Algorithm Track: Use today's systems
Work on developing and testing new quantum algorithms to solve today's problems.

Architecture Track: Improve today's systems
Use computer programs to design and test larger and more accurate quantum hardware systems.

In either track, you can explore what's possible or create value today.

Track Meeting
Full track meets in person to get general guidance (usually Fridays)

Group Meeting
Group of researchers meet in person with no agenda

Group Leader (1 per 20 researchers)
Takes attendance at the weekly group meeting and show up ~5 minutes early

Advisor Meeting
Team and their advisor meet for personal guidance

Team Meeting
Team meets to make progress

Partners
Quantum hardware or quantum applications companies - including Do Quantum Inc.

Initial Interest Activity
Document and video submission of what you learned about an open problem and what you found out you were wrong about

Language Learning Activity
Workshop on the foundations of quantum algorithms or architecture - depending on track

Quantum Algorithms Overview Workshop
Explain the 5 stages of quantum algorithm development. Introduce success criteria, verification, and scaling

Quantum Architecture Overview Workshop
Explain the 4 areas of quantum architechture. Introduce success criteria

Problem Area Activity
Learn and teach your partner the most similar workshop to your problem area - depending on track

Presentation Revision Workshop
Teams give eachother feedback on feasibility and benefit practicality

Hypothesis Creation Workshop
Researchers learn how to phrase a question in an analyzable and falsifiable way / to find truth rather than fake truth - good and bad

Hypothesis Testing Workshop
Researchers learn how people with varying levels of resources learned their work's benefit and feasibility (paper) - good and bad

Research Touchpoint Workshop
Researchers sit in circles of about 15 to help eachother solve challenges and answer questions

Presentation Preperation Workshop
Leadership walks researchers through common mistakes in presentations and practices effective solutions

GitHub/Python Workshop
Teams download VS Code, create a github repository, each create a feature branch, each commit Python changes, push changes, PRs to the main branch, and pulls latest main branch while leadership walk around to support them.

Quantum Job Market Workshop
Leadership shows researchers who's hiring, then researchers searches quantum internships and jobs they'd be qualified for at the end of the program

Paper Publishing Workshop
Teams see examples of good and bad technical and white papers, thereby seeing where to find thiem

LaTeX Workshop
Leadership shows researchers best-practices for formatting their papers using LaTeX

Ask questions ahead of meeting

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Not every team is guaranteed appear advisor

definitely cover differentt ypes of papers

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Semester 1 @ UMD
Foundation and Direction

Week 1
Recruiting

Interest Meeting #1

Week 2
Recruiting

Interest Meeting #2

Pre-Track Meeting Responsibilities:
- Leadership sends emails acceptance emails to researchers
- Leadership invites researchers to platforms

Week 3
Onboarding

Track Meeting | Group Meeting | Advisor Meeting | **Team Meeting**

- **Leadership presents slideshow**
  - accessing platforms (Discord / Drive)
  - setting up discord notifications with researchers
  - sets expectations of culture, program (entr. vs int. vs job), field, problem areas to work on, sponsorship, and drop rate of initial interest activity (expect ~50%)
  - important open questions in each problem area (feasibility / benefits)
- **Leadership explains Initial Interest Activity**
- **Researchers choose problem area and form pairs for Initial Interest Activity**

Pre-Track Meeting Responsibilities:
- Pairs of researchers complete and submit Initial Interest Activity
- Leadership grades Initial Interest Activities
- Leadership notifies researchers of acceptance or rejection
- Leadership updates access to platforms accordingly
- Leadership reserves locations for semester 1 presentations (based on # of likely teams)

Week 4
Kickoff

Track Meeting | Group Meeting | Advisor Meeting | **Team Meeting**

- **Leadership delivers Fundamentals of Quantum Computing workshop to researchers**
- **Leadership presents slideshow**
  - what it means to be a group leader (5 mins early / take attendance)
  - conditions of group leader removal
- **Leadership and group leaders setup weekly group meetings**
  - researchers self-select themselves to be group leaders (schedule / take attendance)
  - group leaders propose weekly day / hour to hold the meeting
  - leadership and group leaders decide locations (available and reservable rooms)
  - researchers choose a weekly group meeting to attend starting next week
- **Leadership explains Language Learning Activity to be done over the next week**
- **Researchers choose problem area and form new pairs for Language Learning Activity**

After the meeting, leadership breifs group leaders on how to take attendance and conditions of demotion

Pre-Track Meeting Responsibilities:
- Researchers optionally ask questions in the discord, no reason to not deliver for partner
- One researcher from each pair walks their partner through the first half of the workshop
- Leadership reserves rooms for group meetings

Week 5
Fundamentals Workshop

Track Meeting | **Group Meeting** | Advisor Meeting | **Team Meeting**

- **The second researcher walks their partner through the second half of the workshop**
- **Leadership delivers their track's overview workshop**
- **Leadership explains Problem Area Activity to be done over the next week**
- **Researchers choose problem area and form new pairs for Problem Area Activity**

All are encouraged to share why this program is useful to them and chat with people who have similar goals

Pre-Track Meeting Responsibilities:
- Researchers optionally ask questions in the discord, no reason to not deliver for partner
- Each researcher learns their individual workshop
- Pairs of researchers meet to teach each other their workshop

Week 6
Hypothesis Creation Workshop

Track Meeting | **Group Meeting** | Advisor Meeting | **Team Meeting**

- **Leadership delivers Hypothesis Creation Workshop to researchers**
- **Leadership explains Hypothesis Creation Activity**
- **Explain that they will start their small project**

quantum learning post meetings?? NEED TO HELP AS THEY HAVE QUANTUM QUESTIONS

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emphasize the long term vision

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Pre-Track Meeting Responsibilities:
- Leadership secures rooms and sets timeframes for quick presentations
- Each researcher chooses or creates a hypothesis they find interesting and submits it
- Each researcher updates their hypothesis submission if it changes
- Each researcher looks up what it's like to test the feasibility and benefits of it
- Leadership sends peer advisor info to researchers

Week 7
Hypothesis Testing Workshop + Team Creation

Track Meeting | **Group Meeting** | Advisor Meeting | **Team Meeting**

- **Leadership delivers Hypothesis Testing Workshop**
- **Leadership advises researchers on how to choose a partner**
  - **Researchers create teams of 2 or 3 total** (area of interest / personality / availability / actions > words)
  - **Researchers update google sheet with team information**
- **Leadership explains expectation before next meeting** (one hypothesis per team / start testing it)
  - **Leadership explains how to reach out to peer advisors to learn about the field**
    - for a meeting
    - optionally throught the program in the discord
  - **Leadership shares presentation signup form**

Types of output:
- hypothessis works out
  - positive data results
- hypothesis doesn't work out
  - survey paper
  - well-documented negative results

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Pre-Track Meeting Responsibilities:
- Each team updates their hypothesis submission if it changes
- Researchers choose day/time/place to present

Week 8
Team Preperation

Track Meeting | **Group Meeting** | Advisor Meeting | **Team Meeting**

- **Leadership delivers overview workshop of the other track (30 mins)**
- **Leadership delivers GitHub workshop (30 mins)**
- **Leadership shares expectations of next week (10 mins) - emphasizing importance of intellectual honesty**
  - What was your approach? (good and bad)
  - What did you learn? (synthesize scientific ideas)
  - What did you actually create? (synthesize scientific ideas)
  - What are you interested in doing next? Why?
- **Leadership delivers Quantum Job Market Workshop in post-meeting** (academia / private industry / public industry / government)

Pre-Track Meeting Responsibilities:
- **Teams present their learning and projects to leadership**
- **Leadership records them and shares them with the cohort**
- **Leadership explains:**
  - Researchers will be able to switch their track, choose their next problem area, and form new teams
  - Why
- Researchers look at others' recordings
- Researchers update their google sheet to reflect their next track and problem area

- **Advisor Guidance**:
  - Understand your team's project
  - Introduce yourself, share structured advice on what they may find interesting in the future based on what you know about the field

Week 9
Team Presentations + Start Project 2

Track Meeting | Group Meeting | **Advisor Meeting** | **Team Meeting**

- **Researchers show up to the track meeting they choose to participate in next** (scare them away from doing it but allow it)
- **Researchers spend time getting to know everyone**
- **Researchers form pairs and can take any topic or work their team created with them** (area of interest / personality / availability / judge on actions not words)
  - 3 is only if the team really wants it, 4 must wait until team merging next week
  - will give credit in publication if others created something
- **Leadership explains expectations for next meeting (team hypothesis and update google sheet)**

Teams meet with at least one peer advisor for advice

white papers to publish

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improve advisor outreach system

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keep track of people who switch tracks

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Peer advisors may select teams before official rankings

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Pre-Track Meeting Responsibilities:
- Researchers optionally merge teams and update google sheet with team information (can bring team hypothesis or work / will give individuals credit if used for publication)
  - If someone is left alone, they can come to the track meeting. If no one wants to team with them they'll be removed from platforms and program
  - ("if you think of an unaccounted for edge case Evren is all ears")
- Leadership secures rooms and sets timeframes for quick presentations

Week 10
Final Team Formation + Presentation Preparation

Track Meeting | **Group Meeting** | Advisor Meeting | **Team Meeting**

- **Researchers choose to merge teams or not** (optional / same teammate criteria / max 4 teammates) and update their google sheet with new team information
- Leadership delivers Presentation Expectations
- **Leadership explains expectations before next meeting**
  - Team signs up for presentation and a team to watch two teams
  - Researchers concisely organize their team's information for leadership, peer advisors, and sponsors to see
  - Researchers reachout to new peer advisors to present and get feedback on feasibility and benefit

Pre-Track Meeting Responsibilities:
- Leadership removes anyone who's left alone after swaps
- Leadership sends team info to partners (if any)
- Researchers choose day/time/place to present

- **Advisor Guidance**:
  - Let your team present their hypothesis and progress on feasibility and benefit to you
  - Introduce yourself, share structured advice on what they may be missing

Week 11
Presentation Revision

Track Meeting | **Group Meeting** | Advisor Meeting | **Team Meeting**

- **Leadership encourages everyone to chat with an advisor this or next week**
- **Teams do Presentation Revision Workshop**

Pre-Track Meeting Responsibilities:
- Leadership sends presentation team info to peer advisors - showing up increases your chance of being with your team (researchers already have peer advisor info)
  - exclude any peer advisors that have proactively paired up with teams already
  - if they have, teams should have updated the google sheet
- researchers rank order peer advisors they prefer, vice verse
- Leadership assigns peers to give feedback during track meeting so each team has >= 1 peer

improve advisor outreach system

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Week 12
Presentations and Pictures

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership attends each track's presentations**
- **Leadership synthesizes feedback and shares it with each team** (discord team chat)
- **During the meeting**
  - **Researchers give feedback (leadership if needed)**
  - **Leadership explains how advisors will be matched and answers any questions**

Teams meet with atleast one new peer advisor for advice

Week 13
Thanksgiving Break

Week 14
Pre-Finals Week

Pre-Track Meeting Responsibilities:
- Leadership matches up researchers with a peer advisor
- Researchers initiate introduction and schedule first advisor meeting
- Leadership shares improvements from researchers and of their own with non-profit

Week 15
Program Improvement + Pizza

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership, researcher, and peer advisors enjoy pizza and debrief about the semester**
- **Leadership learns problems or improvements researchers would like to see in the program**

- **Agenda:**
  - Discuss problems or improvements you'd like to see in the program
  - Group leader submits anonymous summary to leadership

Semester 2 @ UMD
Connections

Pre-Track Meeting Responsibilities:
- Leadership lists big quantum news that happened over the break
- Researchers submit name and question they want aimed at the entire track
- Leadership shares improvements from researchers and of their own with non-profit

Week 16
Track Chat

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership shares big quantum news that happened over the break**
- **Leadership shares any submitted questions and who asked them, for peer support**
- **Leadership reminds everyone that researchers and advisors from all chapters are in the discord, and how to reach out**
- **All teams in a the track get familiar with eachother**

Types of output:
- hypothesis works out
  - positive data results
- hypothesis doesn't work out
  - survey paper
  - well-documented negative results

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Pre-Track Meeting Responsibilities:
- Leadership secures industry speaker to do a talk or workshop

Week 17
Industry Day

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership announces teams that partners will collaborate with**
- **Expert in industry talks about the field and answers questions**
- **Leadership reminds researchers that the final stretch of this program is about truth finding**

Both tracks?

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Pre-Track Meeting Responsibilities:
- Leadership secures academic speaker to do a talk or workshop
- Leadership knows sponsored teams and other chapter's track meeting day/time/google meet
- Leadership secures outstanding team to share what they've learned to another chapter
- Outstanding team shares what they've learned to another chapter

Week 18
Academia Day

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Expert in academia talks about the field and answers questions**

Post-doc talk

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Both tracks?

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Pre-Track Meeting Responsibilities:
- Leadership secures government speaker to do a talk or workshop

Week 19
Government Day

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Expert in academia talks about the field and answers questions**

Both tracks?

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Pre-Track Meeting Responsibilities:
- Leadership chooses a room
- Leadership ensures researchers know which room to go to

Week 20
Cohort Day

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Algorithm and Architecture track merge**
- **Teams do lightning talks to persuade eveyone of the promise of their project** (synthesisze, simple, persuade)
- **Researchers rate each team on a scale 1-5**
- **Team with the highest rating will be selected to present progress to another chapter** (if not accepted, then next highest rated team)

Week 21
Local Opportunities + Other Chapter Preentation

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership explains to researchers how they can leverage their experience in the program on campus and ask researchers if anyone knows of anything else**
- **Outstanding team from another chapter presents their progress virtually and answers questions**
- **Researchers use remaining time to chat with other teams**

Week 22
Spring Break

Pre-Track Meeting Responsibilities:
- Researchers submit name and question they want aimed at the entire track
- Researchers submit a question per team they want help answering that another team may be able to answer

Week 23
Cohort Chat

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership shares any submitted questions and who asked them**
- **Each team pick a question they can answer best**
- **Leadership splits cohort into two groups and questions are answered**
  - **First half spent getting group 1's questions answered**
  - **Second half answering group 2's questions**
  - **If any teams are left over, pick a second team's question to help answer**

WHAT IF PRESENTING AT DIFF. SCHOOL WAS A COHORT ACTIVITY IN PERSON?

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Week 24
Technical and White Paper Workshop

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership Delivers Paper Publishing Workshop**

definitely cover differentt ypes of papers

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Week 25
LaTeX Workshop

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Leadership Delivers LaTeX Workshop**

add reminders for upcoming weeks and better instructions for leadership and researchers

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Pre-Track Meeting Responsibilities:
- Leadership asks teams which will give quick lightnight talks to ensure they have enough time during the track meeting
- Leadership orders presentations in advance
- Researchers submit name and question they want aimed at the entire track
- Leadership shares improvements from researchers and of their own with non-profit

Week 26
Sympoisum Voting + Cohort Chat

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **Algorithm and Architecture track merge**
- **Teams that want to give talks at sympoisum give lightning talks to persuade the cohort to represent them at symposium**
- **Researchers rate each team on a scale 1-10**
- **Teams with the highest ratings give talks (that represent the cohort) at the symposium**
  - limited to number of available spots
- **Leadership shares any submitted questions and who asked them, for peer support**

Pre-Track Meeting Responsibilities:
- Leadership notifies track that the track meeting will be longer than usual
- Leadership secures space and food for sympoisum
- Leadership invites academic and industry experts to symposium
- Leadership invites teams to present on stage or not
  - If invited, teams prepare presentations
- Every team creates a poster
- Leadership prints it and secures a poster board/stand or place to secure it

Week 27
Research Symposium Rehearsal + Cohort Chat

Track Meeting | **Group Meeting** | **Advisor Meeting** | **Team Meeting**

- **All leadership and researchers walk from usual track meeting room to symposium room**
- **Presenters practice being on stage and presenting**

- **Agenda:**
  - Discuss problems or improvements you'd like to see in the program
  - Group leader submits anonymous summary to leadership

GET LASERS AND STONES

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Week 28
Research Symposium

Research Symposium

Pre-Track Meeting Responsibilities:
- Leadership shares improvements from researchers and of their own with non-profit

Week 29
Pre-Finals Week

If we keep second semester more open ended for track meetings, then t2 researchers will merge with t1 researchers who made it through

Add t2 track leads and researchers

.5: Learn | **1: Do, Improve** | **1.5: Apply** | **2: Interview**